System Design Basics

Lesson 4: Reliability and Fault Tolerance

Reliability and Fault Tolerance - Study Notes

Key Concepts

- Redundancy and replication
- Retry logic and backoff
- Circuit breaker pattern
- Graceful degradation

Summary

Reliability ensures a system continues to function despite failures. Redundancy duplicates components so a failure doesn't bring down the whole system. Retry logic with exponential backoff handles transient errors. Circuit breakers detect failures and prevent cascading issues by temporarily halting requests to a failing service. Graceful degradation allows the system to provide limited functionality when parts fail.

Important Points

- Keep retries idempotent.
- Circuit breaker thresholds prevent overload.
- Monitor health endpoints for proactive failure detection.

Example

Implement a microservice that calls an external API with a retry policy and a circuit breaker; if the API fails, return cached data.

Quick Review

- Why use exponential backoff?
- What is graceful degradation?